Claus Horn, Ph.D.

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Profile

Accomplished Data Science and AI leader with over a decade of experience leading teams to build and scale ML-driven platforms and products with applications in life sciences, healthcare, retail, and finance.

Research Interests

As founder of EurekaLabs.ch, I am building a multi-agent AI Scientist platform that integrates automated ideation, interpretable AI, and AI safety to transform and accelerate scientific discovery.

My research in computational biology focuses on developing AI-driven tools for protein design to enable transformative advances in enzyme engineering and the development of novel inhibitors and antibodies. Recent breakthroughs include:

- Solving the decade-old data-leakage problem in protein-ligand data, and building the first AI model that generalizes effectively in binding affinity prediction (accepted at Nature Machine Intelligence, link-to-paper).
- Enabling efficient and scalable optimization of protein function through a novel discrete gradient ascent algorithm (link-to-paper).

Leadership Experience

Team Leadership & People Development

- Led research teams of up to 6 members in academia and industry, including scientists, engineers, and students.
- Supervised over 10 master's theses and PhD projects on topics such as protein optimization with deep RL, NLP for healthcare, and real-world AI applications.
- Provided mentorship across disciplines and career levels, with mentees moving on to ETH, Stanford, Google, and AI-focused startups.

Scientific Strategy & Research Direction

- Defined and executed interdisciplinary research agendas, leading to successful acquisition of $> \$500k$ in competitive funding.
- Directed AI projects combining reinforcement learning, protein language models, and physics-based simulations for protein design.
- Developed industry research roadmaps, ensuring alignment with business needs and innovation goals.
- Initiated and led multi-year, multi-institutional projects funded by SNF, Innosuisse, and armasuisse, including collaborations with Yale, ETH Zurich, and industry partners.

Organizational & Program Leadership

- Directed the AI Extension School (2021–2024) at ZHAW, building an interdisciplinary curriculum for life science professionals transitioning to AI.
- Co-founded SmartElf.ai, an AI startup focused on building a platform for AI agent development, 2023.
- Built the first data science team at Migros, Switzerland's largest retailer, and created MIDEX, the system behind $7B/year in promotion steering.

Community Building & Outreach

- Founded the Autonomous Agents and RL Community (> 3000 members), connecting researchers and

engineers across academia and industry to advance real-world agent applications.
- Actively engaged in science communication and policy advising (e.g., *AI strategy panel for the Swiss Federal Council*).

Formal Leadership Training
- Completed the CAS in Management and Leadership (2022–2023) at HWZ Zurich, covering innovation management, organizational strategy, and people development in tech environments.

Employment

Research Scientist — **New Haven, CT, USA**
Yale University, School of Medicine — *since March 2024*
- Lead AI research in protein modeling and therapeutic discovery.
- Developed predictive models using multimodal biological and clinical data.
- Collaborate with experimentalists and product stakeholders on model-informed design strategies

Advisor - AI & Data Strategy (Part-Time) — **Zurich, Switzerland**
Various Industry Clients (e.g., Kikodo, V-ZUG AG, Bühler Group) — *2018 - 2024*
- Advised startups and enterprise RD teams on designing and deploying agent-based solutions for optimization and decision support.
- Supported AI strategy development and implementation for industrial automation, logistics, and finance.
- Helped translate scientific advances in reinforcement learning and agent-based modeling into production-ready systems.

Director of AI Extension School — **Zurich, Switzerland**
Zurich University of Applied Sciences — *March 2021 - March 2024*
- Led the design, implementation, and strategic growth of a continuing education program in artificial intelligence for professionals, researchers, and industry partners. Oversaw curriculum development, faculty recruitment, and partnerships with leading AI institutions. Directed initiatives to integrate cutting-edge AI research into applied training programs, and managed budget, marketing, operations and program evaluation.

Research Group Leader — **Zurich, Switzerland**
Zurich University of Applied Sciences — *March 2020 - March 2024*
- Built and led a team of 6 scientists and engineers across applied ML projects in medicine, biology, and industry.
- Secured over $400K in competitive funding for cross-disciplinary AI research.

Lead Cognitive Data Science — **Zurich, Switzerland**
Swiss Re — *April 2017 - February 2020*
- Led a data science team focused on NLP, document understanding, and business automation.
- Drove ML product integration in regulatory and underwriting workflows.

Lead Data Scientist — **Zurich, Switzerland**
Migros Group — *January 2014 - March 2017*
- Built the first centralized data science team at Switzerland's largest retailer.
- Created MIDEX, a decision engine managing $7B/year in promotional spending.

Researcher and Strategist — **Zug, Switzerland**
Swiss Hedge Fund — *September 2010 - December 2013*
- Designed ML-driven trading models for high-frequency and systematic strategies.

Postdoctoral Researcher — **Palo Alto, CA, USA**
Stanford University and CERN — *September 2006 - August 2010*
- Co-led data infrastructure and developed novel ML techniques for particle discovery and experimental analysis.

Education

- **CAS in Management and Leadership** — **HWZ Zurich**
  Organizational strategy, innovation, and team development in tech settings — *October 2022 - March 2023*
- **Ph.D. in Physics** — **DESY & University of Hamburg**
  Thesis: Machine learning in particle physics — *April 2003 - July 2006*
- **M.Sc. in Physics** — **RWTH Aachen**
  Top 5% of class; exchange fellowships in Paris and Madrid — *March 2003*

Technical Expertise in AI and Machine Learning

- Applied ML: Built models for protein-ligand binding, patient engagement, promotion response, and NLP-driven document intelligence.
- Modeling Approaches: Proficient in supervised learning, representation learning, graph neural networks, and reinforcement learning.
- Tools & Languages: Python, PyTorch, JAX, Hugging Face, Ray, SQL, Docker, GCP, DVC.
- Deployment: Experienced in end-to-end ML pipelines, from exploratory data analysis to API-based deployment in production.

Honors and Awards

- Excellent Reviewer Award – KDD (2024)
- DIZH Fellowship ($250k) – Zurich (2023)
- Contributed to searches for the Higgs boson. Its discovery was awarded with the **Nobel Prize in Physics**.
- National and international research fellowships (DAAD, KTH Stockholm, TIT Tokyo)
- German Academic Scholarship Foundation – Top 1% of students are selected across disciplines